Mohammed Azharudeen Farook Deen

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About

Robotics Engineer with a focus on **Deep Robot Learning** and **Optimal Control** to build intelligent and safe robots.

Skills

Programming: Python, C++, MATLAB
Machine Learning & AI: PyTorch, JAX, TensorFlow, Deep Learning (CNNs, RNNs, Transformers), Reinforcement Learning (PPO, SAC), Imitation Learning (BC, DAgger), Robot Foundation Models(NVIDIA GROOT, pi-05)
Robotics & Simulation: ROS, ROS2, Gazebo, Isaac Sim, Isaac Gym, Mujoco, NVIDIA Warp
Optimization & Control: Model Predictive Control (MPC), Quadratic/Nonlinear Programming (QP, SQP)
Development Tools & Systems: Git, Docker, Testing Frameworks (PyTest, GoogleTest), Linux
Languages: English (Business Fluency), German (Beginner), Tamil (Native), Hindi (Beginner)

Work Experience

Intern / Working Student — *Stuttgart, Germany*
Fraunhofer IPA and IFF University of Stuttgart — *July 2024 – Mar 2025*

- Developed a simulation environment in **MuJoCo**, supporting both state- and vision-based inputs (using **ResNet/Vision Transformers (ViT)**, and data augmentation) for a peg-in-hole assembly task with different difficulty presets for **Learning from Demonstrations(LfD)** and **Human-in-the-Loop RL.**
- Implemented and tuned various controller modules; Integrated **teleoperation** modules for keyboard and spacemouse control, collected demonstration data and explored **synthetic data generation**.
- Built a real arm robot **manipulator(UR5e + Robotiq hand-e Gripper)** interface with **Flask–ROS server** for command execution with **actor–learner asynchronous nodes**.
- Adapted both the simulation and real environment to work with real-time online **Reinforcement Learning** (SAC) and **Imitation Learning** (BC, DAgger) pipelines in both **Pytorch** and **JAX**, and benchmarked them using simulation.

Master Thesis - Robot Learning of Quadrupedal Locomotion on Deformable Terrain — *Aachen, Germany*
RWTH Institute for Data Science in Mechanical Engineering and RWTH Institute of Geomechanics and Underground Technology — *Sep 2023 – Mar 2024*

- Integrated a particle-based contact model to simulate the complex behaviors of granular materials in **NVIDIA Isaac Sim**, developed and evaluated an adaptable and robust controller using Reinforcement Learning (PPO) in PyTorch for **quadruped (Unitree A1)** to follow a unified policy in diverse environments with **sim-to-real** additions.
- Preliminary work presented at **1st German Robotics Conference (GRC2025)**, and a related paper is being finalized.

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Student Assistant — *Aachen, Germany*
RWTH Institute for Automatic Control — *May 2023 – June 2024*

- Extended codebase of a unified C++ interface for various **quadratic programming (QP)** to **nonlinear programming (NLP) solvers** and included coupled tools for MPC, NMPC, SQP, auto differentiation, Hessian regularization methods, polynomial interpolation, collocation methods, etc with appropriate tests.
- Involved in the Fernbin project, where my tasks included developing simple simulation for testing control systems in Python and **ROS2** for ship-vessels/ferry that react dynamically to the surroundings.

Research Project — *Aachen, Germany*
Automated and Connected Driving Challenges — RWTH Institute for Automotive Engineering — *Apr 2023 – Aug 2023*

- Enhanced trajectory planning in autonomous vehicles using **Model Predictive Control (MPC)**.

Teaching Assistant — *Aachen, Germany*
RWTH Institute of Geomechanics and Underground Technology — *Apr 2023 – Sep 2023*

- Implemented programming exercises in Python/ROS, conducted tutorial sessions for "Introduction to Robotics" course.

Project / Research Assistant — *Kancheepuram, India*
Microscale Transport Laboratory, Indian Institute of Information Technology (IIITDM)) — *Aug 2019 – Mar 2021*

- Studied predominant microscale phenomena subjected to acoustic fields and published results in various reputed journals.

Education

RWTH Aachen University *Oct 2021 – Dec 2024*
Master of Science in ***Robotic Systems Engineering***

Shiv Nadar University *Aug 2015 - May 2019*
Bachelor of Technology with major in ***Mechanical Engineering****(Specilization: Computational Techniques) and minor in Electronics and Communication Engineering*

Honors & Awards

Winner – Munich Humanoid Manipulation Hackathon (BMW Award) *AI.Factory Lab*
Organized by RoboTUM, BMW Group, Siemens, NVIDIA, KIT, Hugging Face *Sep 2025*

Developed a multi-expert agent system for bi-manual manipulation using **Franka Emika robots**, integrating **ACT models** and LLM orchestration. Our team outperformed all others by solving twice as much as the next best team. Awarded the BMW Best Solution Award and an NVIDIA Jetson AGX Orin Developer Kit.

Full Academic Scholarship, 8 Semesters – Shiv Nadar University *Aug 2015*
Issued by Shiv Nadar University

Completed my engineering program with 100% scholarship across all 8 semesters.

Journal Publications and Accepted Conferences

Rapid Quadrupedal Locomotion on Deformable Terrain - 1st German Robotics Conference March 2025
Mohammed Azharudeen Farook Deen, Omer Kemal Adak, Raul Fuentes
https://www.robotics-institute-germany.de/conference/

Theory of nonlinear acoustic forces acting on inhomogeneous fluids. Journal of Fluid Mechanics. [IF(2022)-3.7] April 2022
Rajendran VK, Jayakumar S,*Azharudeen M*, Subramani K
https://doi.org/10.1017/jfm.2022.257

Heat transfer mechanism driven by acoustic body force under acoustic fields. Physical Review Fluids. [IF(2021)-2.5] July, 2021
Azharudeen M, Kumar V, Pothuri C, Subramani K

A Novel Heat Transfer Mechanism Using Acoustic Waves - Presented at International Heat and Mass Transfer Conference, IIT Roorke Dec 2019
Azharudeen M, Pothuri C, Subramani K
https://doi.org/10.1103/PhysRevFluids.6.073501

Rapid mixing in microchannel using standing bulk acoustic waves. Physics of Fluids. [IF(2020)-3.5] Dec 2019
Azharudeen M, Pothuri C, Subramani K
https://doi.org/10.1017/jfm.2022.257